

Basic Configuration for First-Time Use

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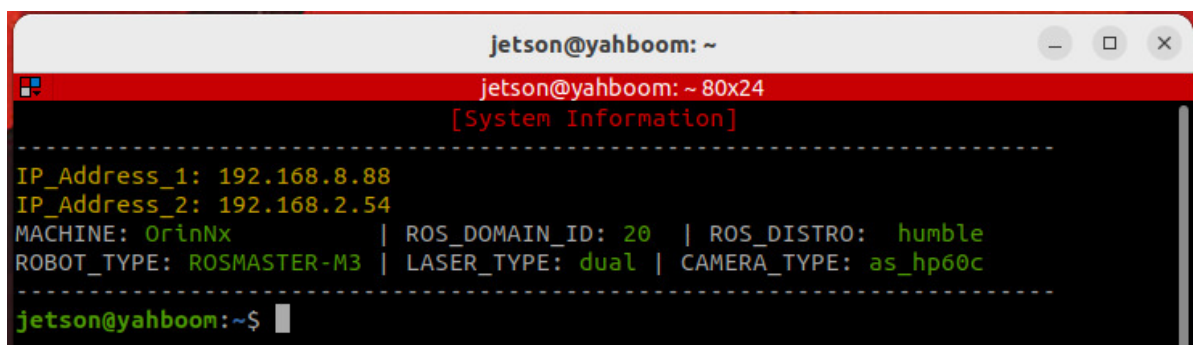
1. Course Content

- Steps required for users to configure the machine for the first time.

2. Configure LiDAR Type

The ROSMASTER-M3 distinguishes between single and dual LiDAR versions. You need to modify the system environment variables according to your car's LiDAR type.

- Open a terminal, and it will print basic system information.

A terminal window titled 'jetson@yahboom: ~' with a red header bar. The terminal displays system information in a structured format with dashed lines separating sections. The information includes IP addresses, machine type, ROS domain ID, ROS distro, robot type, laser type, and camera type.

```
jetson@yahboom: ~  
jetson@yahboom: ~ 80x24  
[System Information]  
-----  
IP_Address_1: 192.168.8.88  
IP_Address_2: 192.168.2.54  
MACHINE: OrinNx | ROS_DOMAIN_ID: 20 | ROS_DISTRO: humble  
ROBOT_TYPE: ROSMASTER-M3 | LASER_TYPE: dual | CAMERA_TYPE: as_hp60c  
-----  
jetson@yahboom:~$
```

Meaning of terminal information:

- `MACHINE`: Car's mainboard type
- `ROBOT_TYPE`: Product robot model
- `ROS_DOMAIN_ID`: ROS domain ID
- `ROS_DISTRO`: ROS version
- `LASER_TYPE`: LiDAR type, distinguishing between single and dual LiDAR versions
- `CAMERA_TYPE`: Distinguishes between Orbbec Dabai_DCW2 and ANSEE Nuwa-HP60C versions

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first:

```
bringup_m3
```

Open a terminal inside the container:

```
exec_m3
```

- Open the `.bashrc` file to modify `LASER_TYPE`:

```
nano ~/.bashrc
```

- Uncomment the corresponding line based on your car's single or dual LiDAR type. The following image shows an example for a single LiDAR version.
- For a single LiDAR version, uncomment `export LASER_TYPE=single` and comment out `#export LASER_TYPE=dual`.
- For a dual LiDAR version, uncomment `export LASER_TYPE=dual` and comment out `#export LASER_TYPE=single`.

```
#=====雷达和相机型号=====
#=====Radar and camera models=====
#=====
#单雷达：single                双雷达：dual
#Single radar: single          Dual radar: dual
export LASER_TYPE=single
#export LASER_TYPE=dual
#=====
#相机类型    dabai_dcw2                as_hp60c
#Camera type dabai_dcw2                as_hp60c
#export CAMERA_TYPE=dabai_dcw2
export CAMERA_TYPE=as_hp60c
#=====
#export MACHINE=OriNaga
```

3. Configure Camera Type

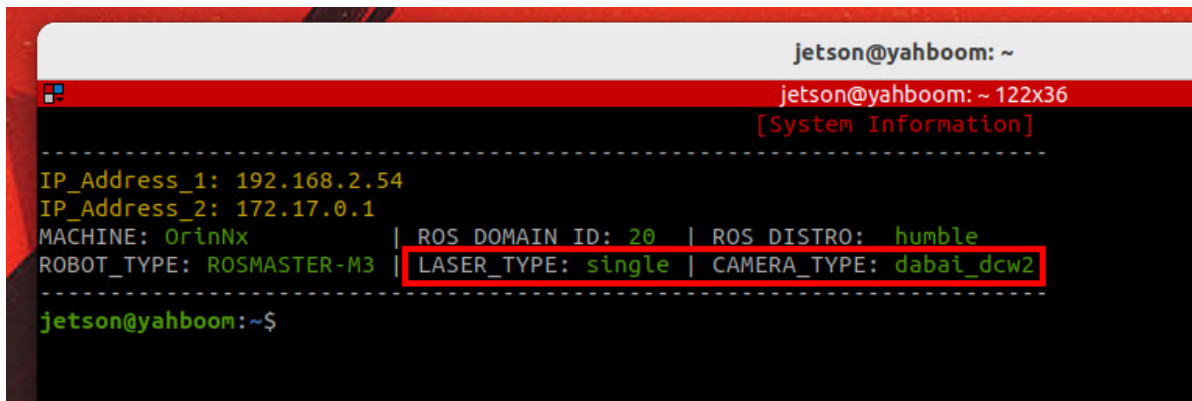
- Similarly, modify `CAMERA_TYPE` to your camera type. The following image shows an example for the Dabai_DCW2 camera.
- Orbbec Dabai_DCW2: `dabai_dcw2`
- ANSEE Nuwa-HP60C: `as_hp60c`

```
#=====雷达和相机型号=====
#=====Radar and camera models=====
#=====
#单雷达：single                双雷达：dual
#Single radar: single          Dual radar: dual
export LASER_TYPE=single
#export LASER_TYPE=dual
#=====
#相机类型    dabai_dcw2                as_hp60c
#Camera type dabai_dcw2                as_hp60c
export CAMERA_TYPE=dabai_dcw2
#export CAMERA_TYPE=as_hp60c
#=====
#export MACHINE=OriNaga
```

- After modifying, press `Ctrl+X`, then enter `y` to save and exit.
- Refresh the environment variables:

```
source ~/.bashrc
```

Check that `LASER_TYPE` and `CAMERA_TYPE` in the terminal information match your LiDAR and camera types.



```
jetson@yahboom: ~  
jetson@yahboom: ~ 122x36  
[System Information]  
-----  
IP_Address_1: 192.168.2.54  
IP_Address_2: 172.17.0.1  
MACHINE: OrinNx | ROS DOMAIN ID: 20 | ROS DISTRO: humble  
ROBOT_TYPE: ROSMASTER-M3 | LASER_TYPE: single | CAMERA_TYPE: dabai_dcw2  
-----  
jetson@yahboom:~$
```

The image shows a terminal window with a red title bar. The prompt is 'jetson@yahboom: ~'. Below it, the terminal displays system information. The line 'ROBOT_TYPE: ROSMASTER-M3 | LASER_TYPE: single | CAMERA_TYPE: dabai_dcw2' is highlighted with a red rectangular box. The prompt at the bottom is 'jetson@yahboom:~\$'.